(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC-205 **Subject: Discrete Mathematics**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Answer the following short answer type questions (Compulsory):-

(a) Define a power set and give an example. **(2)**
(b) State De Morgan's Laws in propositional logic. **(2)**
(c) What is the principle of mathematical induction? **(2)**
(d) Give an example of an equivalence relation. **(2)**
(e) Define a Boolean Algebra. **(2)**
(f) Explain the difference between a group and a monoid. **(2)**
(g) What is Euler's formula in graph theory? **(2)**
(h) Define a Hamiltonian circuit in a graph. **(1)**

UNIT-I

Q2 (a) Construct a Venn diagram for three sets A, B, C with the following conditions: $A \cap B \neq \emptyset, B \cap C \neq \emptyset, A \cap C \neq \emptyset$. **(5)**

(b) Create a truth table for the proposition $p \rightarrow (q \wedge \neg r)$ and determine its logical equivalence. **(5)**

(c) Define a binary relation R on the set {1, 2, 3, 4} where R contains (a, b) if and only if a divides b. Determine if R is reflexive, symmetric, and transitive, providing justification for each property. **(5)**

Q3 (a) Using the principle of inclusion and exclusion, find the number of integers between 1 and 100 that are divisible by 2, 3, or 5. Provide step-by-step calculations. **(5)**

(b) Prove that in a group of six people, at least three people either all know each other or none of them knows the others. Use the pigeonhole principle in your proof. **(5)**

(c) Given a set S of n elements, derive an expression for the number of ways to choose a non-empty subset of S. Validate your expression with n = 5. **(5)**

UNIT-II

Q4 (a) Let f(x) = 2x - 1 and g(x) = x/2 + 1. Find the compositions f(g(x)) and g(f(x)) and determine their domains. Illustrate each step and clarify the reasoning behind the determination of their domains. **(5)**

(b) Show that every finite distributive lattice can be represented as a lattice of sets, using the lattice formed by the subsets of the set {a, b, c} as an example. Provide a detailed explanation. **(5)**

(c) Design a logic circuit for the Boolean expression $\overline{(A+\overline{B})C} + \overline{AC}$ and simplify it using Karnaugh maps. Include a step-by-step process for the circuit and a detailed explanation of the Karnaugh map simplification. **(5)**

P.T.O.

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P-1/2